HYDROWORX

ECO

Owner's Manual



HYDROWORX
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1.0 Disclaimer and Confidentiality Notice

This confidential Instruction Manual (the "Manual") is delivered to parties (the "Recipient") who have purchased a HydroWorx System from HydroWorx International, Inc. This Manual is highly confidential and proprietary. Receipt of the Manual is solely in connection with the Recipient's consideration of the enclosed information. Use of this Manual for any other purpose is not authorized.

By accepting this Manual, the Recipient agrees to keep confidential the information contained herein or made available by the Company in connection with any service, maintenance, or repair of the HydroWorx System. By accepting this Manual, the Recipient also agrees to make no copies of the Manual in whole, in part, or in any form and to use this Manual solely for the purposes of understanding the proper use of the HydroWorx System.

2.0 Product Description

The HydroWorx ECO is a self-contained Hydrotherapy system constructed of a welded polypropylene structure with acrylic windows and doors. It incorporates a treadmill, therapy jet with removable massage hose, variable water depth, a seat, handrails, touch screen controls, and a water recirculation system.

The ECO can accommodate a single user with a maximum weight of 300lb. A therapist may accompany the patient in the unit with a maximum combined weight of 550lb. The therapist should not be on the treadmill belt while the patient is walking/running, instead the therapist should stand on the treadmill side guards.

A unique feature of the ECO is the dual water chamber design. The small chamber at the rear of the unit is referred to as the foyer. The foyer serves as a water equalization chamber to gain entry to the main chamber and also serves as a holding chamber for variable water depth.

Specifications	
Weight (empty)	1500 lb
Weight (full)	5200 lb
Length	107"
Width	34.5"
Height	71"
Volume of water	440 gal
Max water depth	50"
Min water depth(w/ 50" max)	31.5"
Recirculation flow rate	15-35 gpm
Transfer time (to equalize)	1.5 minutes
Transfer time (to fill main chamber)	1.5 minutes
Treadmill speed	.2-6 mph

3.0 Limited Warranty

HydroWorx International, Inc. (HydroWorx) warrants solely to the original Buyer that this product, consisting of the treadmill system, jet system, electronic controls, and water circulation equipment will be free from defects in materials and workmanship, when given normal, proper and intended installation and usage, for the period set forth below from the date noted on signed start-up completion form.

LIMITED WARRANTY PERIOD	COMPONENTS COVERED
One (1) Year	▪ Water Circulation Equipment* (heater, filter housing, erosion feeder and pump) ▪ Therapy Jet and Pump ▪ Treadmill System* (Motor and drive components, bearings, seals, belt) ▪ Control System ▪ HMI System ▪ Water transfer system (pump, valves, actuator) ▪ Door seals
Five (5) Year	▪ Tank Structural Integrity

*Proper maintenance schedules must be followed to keep the warranty intact.

During the limited warranty period, HydroWorx will repair or replace, at its option and without cost to the Buyer, any portion of the product which is confirmed by HydroWorx to be defective.

This warranty shall not apply to any expendable component, such as, but not limited to chemicals, fuses, filters, and lubricants. HydroWorx shall have no obligations to make repairs or replacements which result, in whole or in part, from (I) catastrophe, fault or negligence of Buyer, (II) improper or unauthorized use of product, (III) maintenance or installation of the product which is not in full compliance with the requirements of the HydroWorx Owner's Manual and Design and Installation Guide, (IV) the presence of debris or silt in the product, (V) the failure to maintain water cleanliness and chemistry, accordance with the recommendations of recirculation equipment suppliers, (VI) any modification of the product (including, but not limited to, removal or affixing any object to the equipment or removing

any component part of the product without prior specific written instruction from HydroWorx), or (VII) causes external to the product such as, but not limited to, power failure.

In the event after inspection by HydroWorx the product proves not to be defective or is not subject to the warranty protection because of normal use, wear and tear, misuse, lapse of time, accident, modification, or any other exclusion specified above, the Buyer will be charged for repair or replacement services at HydroWorx's standard rates then in effect together with the "portal to portal" expenses incurred by HydroWorx in providing the service.

The provisions of the foregoing warranty are in lieu of any other warranty, whether express or implied, written or oral, including any warranty of merchantability or fitness for a particular purpose. HydroWorx's liability arising out of the manufacture, sale or supplying of the products or its use or disposition whether based upon warranty, contract, tort or otherwise, shall not exceed the actual purchase price paid by the buyer for the product. In no event shall HydroWorx be liable to buyer or any other person or entity including, but not limited to, loss of profits or loss of use of damages arising out of the manufacture, sale or supplying of the product the foregoing warranty applies to the buyer only and shall not extend to any other person or entity, including, without limitation, customers of the buyer.

Some states do not allow limitations on how long an implied warranty lasts, or the exclusion or limitation of incidental or consequential damages, so the above limitation or exclusion may not apply to some buyers. This warranty gives the buyer specific legal rights, and the buyer may also have other rights which vary from state to state.

Warning: user or patient should not use this equipment without constant adult supervision from a qualified practitioner who has been trained on the use of the HydroWorx system. Patients or users should consult his/her physician before using this equipment. At all times HydroWorx disclaims and shall not be liable for any personal injury sustained by any user whether authorized or unauthorized, of the product.

Any modification to the as-delivered and installed unit that is not made by the factory or factory authorized personnel shall be interpreted as an unauthorized modification. Any unauthorized modification that is not reviewed and approved by HydroWorx prior to making such modification of the unit or any other device sold by or through HydroWorx shall release HydroWorx of any and all liability for personal injury or property damage relating to or arising from use with the modified product. Any modification to a HydroWorx product not approved by HydroWorx shall void any and all warrantee of the product.

The unit must be installed in accordance with the ECO Design & Installation Guide. Any variance not approved by HydroWorx may result in a voided warranty.

4.0 Safety

READ AND FOLLOW ALL SAFETY INSTRUCTIONS.

READ THIS MANUAL ENTIRELY BEFORE OPERATING THE UNIT.

SAVE THESE INSTRUCTIONS.

⚠ DANGER

- **Risk of Injury**
  - Replace damaged power cord immediately.
  - Do not bury power cord.
  - Connect to a grounded, grounding type receptacle only.
- **Risk of Accidental Drowning**. Extreme caution must be exercised to prevent unauthorized access by children. To avoid accidents, ensure that children cannot use this spa unless they are supervised at all times.
- **Risk of Injury.** The suction fittings in this unit are sized to match the specific water flow created by the pump. Should the need arise to replace the suction fittings or the pump, be sure that the flow rates are compatible.
  Never operate the unit if the suction fittings are broken or missing. Never replace a suction fitting with one rated less than the flow rate marked on the original suction fitting.
- **Risk of Electric Shock.** Install at least 5 feet from all metal surfaces. As an alternative, the unit may be installed within 5 feet of metal surfaces if each metal surface is permanently connected by a minimum 8AWG solid copper conductor to the wire connector on the terminal box that is provided for this purpose.
- **Risk of Electric Shock.** Do not permit any electrical appliance, such as a light, telephone, radio, or television, within 5 feet of the unit.

⚠ WARNING

- For indoor use only. This unit is not intended for outdoor use.
- To reduce the risk of injury
  - The water in the unit should never exceed 104°F. Water temperatures between 100°F and 104°F are considered safe for a healthy adult. Lower water temperatures are recommended for young children and when use exceeds 10 minutes.
  - Since excessive water temperatures have a high potential for causing fetal damage during the early months of pregnancy, pregnant or possibly pregnant women should limit the water temperature to 100°F.
  - Before entering the unit, the user should measure the water temperature since the tolerance of water temperature-regulating devices varies.

- The use of alcohol, drugs, or medication before or during use of the unit may lead to unconsciousness with the possibility of drowning.
- Obese persons and persons with a history of heart disease, low or high blood pressure, circulatory system problems, or diabetes should consult a physician before entering the unit.
- Persons using medication should consult a physician before using the unit since some medication may induce drowsiness while other medication may affect heart rate, blood pressure, and circulation.

- **REDUCE THE RISK OF ELECTROCUTION**
  - Be sure spa is protected by a ground fault interrupter breaker and test prior to each use.
- **REDUCE THE RISK OF CHILD DROWNING**
  - To reduce the risk of injury, do not permit children to use this product unless they are closely supervised at all times.
  - Attach spa cover after each use.
- User's shall always have supervision from a qualified aquatic therapy practitioner or exercise specialist (therapist).
- If the user begins to feel unwell while using the unit it is important to stop the session immediately. It takes approximately 3 minutes to exit the unit from maximum depth.

⚠ CAUTION

- Unauthorized service, repair, or replacement of parts may void your warranty and may damage the unit or injure the user or therapist.
- Do not climb over the sides of the unit unless in the event of an emergency. This could damage the unit or cause a fall.
- Do not open the foyer door with water in the unit unless in the event of an emergency. The force of the rushing water could cause someone to fall and could damage the door.
- Maintain proper water chemistry. Improper water chemistry may cause skin and eye irritation as well as damage to the unit.

SAFETY FEATURES

- The unit is equipped with an emergency-stop button located on the front cabinet doors. In the event of an emergency, pressing this button will disable power to the unit.
- The unit is equipped with an all-stop button located below the HMI screen. When pressed this button will turn off the jet, stop the treadmill, and begin transferring water to the foyer (equalizing).

- The unit is equipped with a safety tether. The user should wear this on their wrist. In the event of a fall, the cord can be pulled and result in an all-stop function as described above.
- The unit is equipped with GFCI protection.
- Installation shall comply with this document, the ECO Design and Installation Guide, the National Electric Code, and any other local codes.
- A wire connector is provided on this unit to connect a minimum 8AWG solid copper conductor between this unit and any metal equipment, metal enclosures or electrical equipment, metal water pipe, or conduit within 5 feet of the unit.
- Prior to filling the unit, remove the temporary label located inside the main chamber. Also prior to filling the unit, install the provided hydrotherapy room warning sign within line of sight of the unit.

5.0 Operating Instructions

Refer to Appendix A for diagrams showing the various components of the ECO.

5.1 Dual Water Chambers

A unique feature of the ECO is the dual water chamber design. The small chamber at the rear of the unit is referred to as the foyer. The foyer serves as a water equalization chamber to gain entry to the main chamber and also serves as a holding chamber for variable water depth.

When the unit is not in use all the water is held in the main chamber where it is heated, filtered, and sanitized 24/7. If the unit has not been used in several hours, it is recommended to run a full transfer cycle and run the jet to replace any cool water that remained in the plumbing.

To enter the main chamber some of the water will be transferred to the foyer until the water height is equal between the two chambers. The command to equalize the water can be given in either of two ways. The main method is to press the transfer button located to the hinge side of the main door as shown in Appendix A.
The other method is to press the stop session button on the HMI screen. When the command is given a valve under the foyer floor opens allowing the water to equalize.

When the water equalizes, the main door can be opened, the patient can enter the main chamber and then close the main door behind him/her.

To transfer water from the foyer to the main chamber the command can be given in either of two ways. The main method is to press the level up arrow on the HMI screen. When a new session is started the HMI will prompt the user to verify the main door is closed and then the user will have to press the level up arrow again.

The other method is to press the transfer button located at the hinge side of the main door as shown in Appendix A. When the command is given the valve under the foyer floor closes and a pump turns on and transfers the water from the foyer to the main chamber. If you don't want to fill the main chamber to the max water depth of 50" you can press the level stop button on the HMI screen.

To achieve variable water depth during the session, use the level buttons on the HMI screen. To change water depth, press the level up or down arrows and when the desired depth is achieved, press the level stop button. Adjusting the water level varies the amount of body weight supported by the patient's lower body.

To exit the main chamber the water must first be equalized so the patient can move to the foyer and then the foyer must be emptied to exit the unit. To initiate the first step, press the stop session button on the HMI screen. The water can also be equalized via the transfer button, but the stop session is recommended because it resets the data tracker for the next session. Once the water equalizes, the patient can move to the foyer. To exit the foyer, press the transfer button and wait for the foyer to empty.

5.2 Doors

5.2.1 Foyer Door

To open the foyer door:

1. ⚠ CAUTION **Ensure all the water has transferred from the foyer before opening the door.**
2. Lift door latch handle over the catch.
3. Rotate door latch counterclockwise.
4. Swing the door open.

To close the foyer door:

1. Verify the door latch is open. It should be rotated all the way counterclockwise.
2. Swing the door closed.
3. Hold the door closed while closing the door latch by rotating it clockwise.
   ⚠ CAUTION **Beware of the pinch point between the door latch and the door.**
4. Lift door latch handle over the catch and release it so that the door latch is securely captured.

5.2.2 Main Door

To open the main door:

1. Ensure the water has equalized before opening the door.
2. Lift door latch handle over the catch.
3. Rotate door latch clockwise.
4. Swing the door open.

To close the main door:

1. Verify the door latch is open. It should be rotated all the way clockwise.
2. Swing the door closed.
3. Hold the door closed while closing the door latch by rotating it counterclockwise.
   ⚠ CAUTION **Beware of the pinch point between the door latch and the door.**
4. Lift door latch handle over the catch and release it so that the door latch is securely captured.

5.3 Treadmill

The ECO is equipped with a variable speed treadmill. By adjusting the speed, the therapist can vary the intensity of the session. The patient should stay centered left to right on the treadmill and not put any sideways force on the belt that could cause a belt tracking problem.

To operate the treadmill:

1. Verify the main door is closed before starting the treadmill.
2. It is recommended that the patient straddle the treadmill by placing one foot on each side of the treadmill, onto the side guards before turning on the treadmill.
3. To activate the treadmill, press the speed on/off button on the HMI screen. The treadmill will not start moving until a speed is selected.
4. On the HMI screen either select a preset speed or press the speed increase button. The treadmill has a ramp up function so it may take a couple seconds to reach the desired speed. The user can change the range of preset speeds by holding the mode button on the HMI screen to toggle between sport and therapy mode.
5. ⚠ CAUTION It is recommended to let one foot touch the treadmill belt to get a feeling for how fast the patient must walk/run before stepping onto the treadmill with both feet.
6. During the session speed may be increased or decreased by either selecting a preset speed or by pressing the speed increase and decrease buttons on the HMI screen.
7. The speed increase and decrease buttons on the HMI screen adjust the speed in 0.1mph increments each time they are pressed. If the user holds the button, then the speed will continue to change as long as the button is held. The speed can be increased/decreased outside of the preset range by using the speed increase and decrease buttons.
8. To stop the treadmill there are several ways to do so depending on the reason why the treadmill is to be stopped.
   a. Speed on/off button on the HMI screen: this is the preferred method if the user just wants to stop the treadmill but continue the session.
   b. Pause Session button on the HMI screen: this is the preferred method if the user wants to stop the treadmill and jet but continue the session.

c. Stop Session button on the HMI screen: use this button when the session is over. This button will stop the treadmill and jet, begin to equalize the water, and clear the session timer and distance tracker.

9. The treadmill is programmed to automatically shut off after 1 hour of continuous use. If an extended session is desired, the treadmill can be restarted immediately.

5.4 Therapy Jet

The ECO includes a therapy jet that can be used as resistance for running and upper body exercises, or it can be used with the massage hose for muscle relaxation.

Turning the Therapy Jet on and off:

1. If the water level is at max depth press the jet button on the HMI.
2. If the water level is not a max depth hold the jet button on the HMI for 3 seconds.
3. Press the jet button on the HMI to turn the jet off.
4. The jet is programmed to automatically shut off after 1 hour of continuous use. If an extended session is desired, the jet can be restarted immediately.

The flow rate and direction can be varied. To adjust the flow rate, rotate the nozzle located in the center of the jet. Turning the nozzle clockwise will decrease the flow rate. Turning the nozzle counterclockwise will increase the flow rate. To adjust the flow direction, grasp the nozzle and push in the desired direction. The nozzle may be difficult to move at first.

To use the massage hose:

1. Make sure jet is off before connecting or disconnecting.
2. Grab the end of the massage hose with the clip-on device.
3. Pull back on the clip and hold while pushing onto the jet nozzle.
4. When fully inserted onto the jet nozzle, push outside ring of the clip toward the jet to lock the massage hose in place.
5. **NOTICE** Point the end of the hose down into the water to avoid water spraying out of the unit.
6. Turn on the jet by pressing the jet button on the HMI screen

5.5 Heater

The ECO is equipped with an Electric Heater. The water temperature can be adjusted by turning the dial on the heater to the desired temperature. Recommended water temperature is 90°F. It can take 1.5 days for the water to heat up when the unit is filled with cold tap water.

The ECO is equipped with a digital thermometer. The body of the thermometer can be suction cupped to the unit and the thermocouple placed in the water. Check the temperature of the water before each use. If the temperature is higher than 104°F do not enter until the

temperature drops below 104°F. If the temperature does not drop below 104°F, then turn off power to the unit and contact HydroWorx.

⚠ WARNING Prolonged immersion in hot water may induce hyperthermia. Hyperthermia occurs when the internal temperature of the body reaches a level several degrees above the normal body temperature of 98.6°F. The symptoms of hyperthermia include an increase in the internal temperature of the body, dizziness, lethargy, drowsiness, and fainting. The effects of hyperthermia include failure to perceive heat; failure to recognize the need to exit the unit; unawareness of impending hazard; fetal damage in pregnant women; physical inability to exit the unit; and unconsciousness resulting in the danger of drowning. The use of alcohol, drugs, or medication can greatly increase the risk of fatal hyperthermia.

5.6 Seat

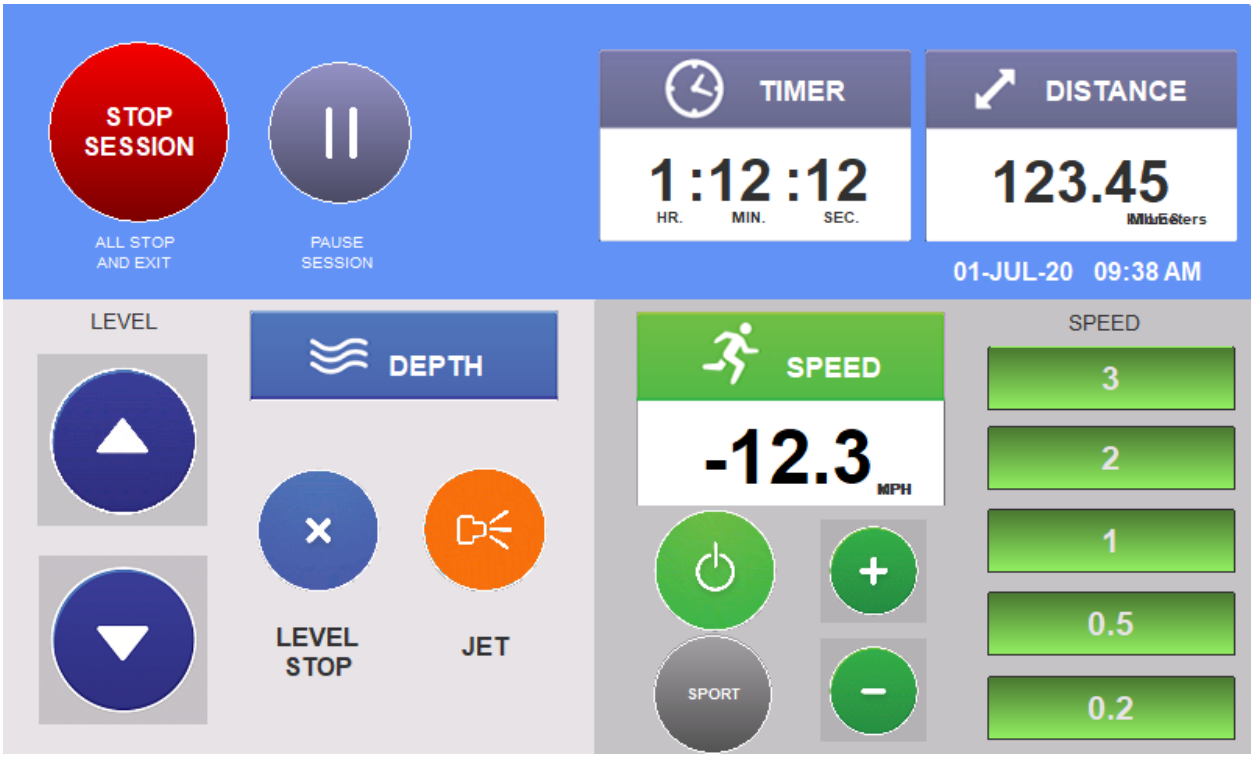
The ECO is equipped with a fold-up seat in the main chamber. The seat can be used for various exercises like lower body stretches or a place to rest while doing upper body exercises. Choose a suitable water level when using the seat. **⚠ CAUTION** The maximum water level is too high for most people.

5.7 Handholds

The ECO is equipped with handholds in the foyer and main chamber. The handholds provide stability when entering and exiting as well as when exercising.

5.8 HMI

The ECO is equipped with an HMI screen (Human Machine Interface). The screen is touchscreen and works with wet fingers.



- Stop Session: When pressed this button will turn off the jet, stop the treadmill, and begin transferring water to the foyer(equalizing).
- Pause Session: When pressed this button will stop the treadmill and jet but continue the session.
- Timer: The timer displays the run time of the session. The timer begins when the treadmill or jet is turned on. The timer is paused if the pause session button is pressed. The timer can be manually reset by holding the pause session button for 5 seconds. The timer will automatically reset when a new session begins. (The stop session button must have been pressed at the end of the previous session)
- Distance: This display shows the distance travelled on the treadmill during a session. The distance can be manually reset by holding the pause session button for 5 seconds. The distance will automatically reset when a new session begins. (The stop session button must have been pressed at the end of the previous session)
- Date and Time: This displays the local date and time.
- Speed: The speed section of the HMI screen shows the speed of the treadmill, turns the treadmill on and off and adjusts speed of the treadmill.
    - The power button turns the treadmill on and off. The treadmill will not start until a speed is selected.
    - The + and – buttons increase or decrease the speed in 0.1mph increments. If a button is held the speed will increase/decrease until the button is released.
    - There are pre-set speeds along the right side for convenience.

- The Sport/Therapy mode button if held will switch between sport and therapy modes. The mode changes the pre-set speeds.

- Level/Depth: This section controls the water depth of the main chamber
  - Press the arrow up and down to initiate a change in water level.
  - Press the level stop button when the desired depth is achieved.
  - Water transfer will automatically stop at upper and lower limits.
- Jet: When pressed this button will turn the jet on and off. A single press will turn the jet on if the water level is at max depth. If the water level is not a max depth hold the button for 3 seconds.

6.0 Maintenance Instructions

6.1 Maintenance quick reference schedule

Daily:

- ✓ Check bromine and pH levels.

Weekly:

- ✓ Check bromine tablet level in the erosion feeder.
- ✓ Shock water with lithium hypochlorite shock.
- ✓ Check alkalinity and calcium hardness levels.
- ✓ Check recirculation flow rate.
- ✓ Refill unit.
- ✓ Wipe down the unit.
- ✓ Check treadmill tracking and tension

Monthly:

- ✓ Test the GFCI devices.
- ✓ Drain and thoroughly clean the unit.

6.2 Water Maintenance

Correct water quality is critical for patient health and comfort as well as longevity of the unit. The ECO is supplied with a Taylor water chemistry test kit. This test kit will be used to determine chemical levels and recommendations on how to correct those levels if needed.

6.2.1 Chemicals needed to maintain the water:

- Alkalinity Increaser
- Alkalinity Decreaser-Dry Acid most recommended or Muriatic Acid
- Calcium Hardness Increaser
- Lithium Hypochlorite based Shock
- pH Increaser

- pH Decreaser
- 1" Bromine Tablets
- Chem-Out Chlorine reducer/neutralizer

6.2.2 Potential Issues with Imbalanced Water

- Low Bromine: Inadequate disinfection. Does not kill contaminants properly.
- High Bromine: Corrodes surfaces and equipment. Irritates/ burns user's skin, eyes and respiratory system.
- Low pH: Corrodes surfaces & equipment. Excess sanitizer use. Irritates bather's skin & eyes. Plugs filter. Decrease your sanitizer effectiveness.
- High pH: Scale deposits on surfaces & equipment. May contribute to cloudy water. Poor sanitizer efficiency. Eye and skin irritation. Stains surfaces.
- Low Alkalinity: pH difficult to maintain. Corrosion tendencies. Stains surfaces. Dissolves metals.
- High Alkalinity: pH difficult to adjust, it remains fixed. Cloudy water. Potential for scaling. High pH and low sanitizer efficiency, sanitizer is pH dependent. Potential to plug filter.
- Low Calcium Hardness: May lead to equipment corrosion.
- High Calcium Hardness: May contribute to cloudy water. Scaling of surface, piping, and equipment. Can cause heater inefficiency. Plugs filters. Forms a white film.

6.2.3 Alkalinity

Alkalinity is your pH buffer. It protects your pH from rapid fluctuations. If any of your water chemistry needs adjusting always adjust your alkalinity first. Total alkalinity is the measure of the ability of water to resist changes in pH. Your alkalinity helps level your pH. Often leveling your Alkalinity will balance your pH. Note: You cannot add more than one pound of alkalinity decreaser a day.

Ideal Range (80 to 120 ppm)
Safe Ranges from National Swimming Pool Foundations (60-180 ppm)
Weekly Test Required

6.2.4 Calcium Hardness

Water that contains little to no magnesium is called soft. If you have soft water when you add chemicals into the water, there is nothing for them to hold onto and they won't be effective. There are no chemicals that you can add to decrease your hard water. If your water is too hard you will need to use a water softener when filling the unit.

Recommend Range (250 ppm–350 ppm)
Weekly Test Required

6.2.5 pH

pH is very volatile. Minor issues can greatly change your pH. If your pH is low, then it is acidic. If it is high, then it is basic. Alkalinity and pH work hand in hand with each other. If one is out of range often the other will follow. Having your pH and Alkalinity in safe ranges is going to increase how effective your sanitizer will be.

Ideal Range (7.2 to 7.6)
Safe Ranges from National Swimming Pool Foundations (pH-7.2-7.8)
Daily Test Required

6.2.6 Bromine

Bromine is the primary sanitizer for the ECO. To control the Bromine level, adjust the feeder dial on the erosion feeder and the amount of bromine tablets in the erosion feeder. Slightly increasing your dial will increase the bromine but this does take up to 24 hours before the increase is noticed. Increasing the Bromine level is much easier then decreasing the bromine level. The recommended initial setting is to set the dial on the erosion feeder to .5 and put 5 1" tablets in the feeder. 5 tablets should last 1 week under low usage. The higher the patient load, the more bromine will be needed so adjust the feeder dial and tablets as needed.

Adding tablets to the erosion feeder process:

1. The feeder can be filled with the recirculation pump on.
2. Turn the dial on the erosion feeder to 0. Make note of where the dial was set so you can return it to the same setting after your done adding tablets.
3. Unscrew the lid on the erosion feeder. ⚠ CAUTION Do not inhale fumes.
4. Add desired number of tablets. Some water may spill out so use a paper towel to dry it.
5. Screw the lid back on the erosion feeder.
6. Turn the dial back to the desired setting.

Ideal Range (3.0 to 5.0 ppm)
Safe Ranges from National Swimming Pool Foundations (2-10 ppm)
Daily Test Required

6.2.7 Filter

The ECO is equipped with a cartridge filter and spare cartridge. The filter cartridges are cleanable and reusable. To monitor when the filter needs cleaned, look at the pressure gauge on top of the filter. When the pressure increases above 8psi the filter needs cleaned. If the filter pressure increases too much it will restrict the flow to the heater potentially causing damage to the heater or it may not heat correctly.

Filter cleaning process:

7. Equalize water between the foyer and the main chamber.
8. Turn off power to the recirculation pump. This can be done by switching the breaker off in the ECO's load center or by turning off the main power to the ECO.
9. Raise the lid on the top of the equipment cabinet.
10. Unscrew the lid on the filter and remove the filter cartridge.

11. Rinse the inside of the filter and vacuum the water out using a wet/dry vac.
12. Install a clean filter cartridge and screw the lid back on the filter.
13. Transfer the water back to the main chamber.
14. Turn the power back on to the recirculation pump. If the pump does not prime right away, turn it off for 5 seconds and then turn it back on.
15. Partially unscrew the bleeder on top of the filter. This allows the air out of the system.
16. When a steady flow of water is coming out of the bleeder, the bleeder can be screwed closed.
17. Bleed the air out of the erosion feeder by unscrewing the lid until water reaches the top of the feeder.
18. The unit is now operational again.
19. To clean the used filter cartridge, soak in a tub with a pool filter cleaning solution such as Filter Cleanse. Follow cleaner manufacturer's directions.

6.3 Cleaning

The ECO should be wiped down weekly and thoroughly cleaned monthly. The ECO is made of polypropylene, acrylic, and stainless steel. These materials are corrosion resistant but proper cleaning is needed to maintain a like new appearance.

6.3.1 What to use:

- Mild detergent such as dish soap
- Plastic cleaner such as Novus #1 or Brillianize
- Clean, soft cotton or microfiber cloth
- 10-20% vinegar solution for scale buildup when needed
- Stainless steel cleaner such as Spectra-Clean system 1, 3M, and Weiman

6.3.2 What not to use:

- Scour pads, scotch-brite pads, steel wool, sandpaper
- Paper towels
- Ammonia based cleaners like multipurpose kitchen/bathroom cleaners or Windex
- Alcohol based cleaners
- Bleach

6.3.3 Weekly cleaning

Every week the unit should be wiped down both inside and out. Equalize the water between the foyer and the main chamber. Use a plastic cleaner or mild detergent to clean plastics and a stainless steel cleaner to clean the metal handrails, door latches, and transfer valve actuator rod. If the stainless steel starts to stain, then it should be cleaned more often.

6.3.4 Monthly Cleaning

Every month the unit should be drained, thoroughly cleaned, and refilled with fresh water. The frequency of this routine cleaning can be adjusted based on patient load.

1. First equalize the water between the foyer and the main chamber.
2. Turn power off to the unit.
3. Drain the unit by either opening the drain valve located at the lower front left corner of the unit or use a sump pump.
4. Remove the foyer floor plates and treadmill guards.
5. Pick up the rear of the treadmill and flip it forward so it leans against the front wall inside the unit. Spring loaded keepers will fold out and extend over the side handrails to keep the treadmill in the raised position while the unit is being cleaned.
6. Use a wet/dry vac and/or a sump pump to remove the rest of the water from the bottom of the unit.
7. Thoroughly clean the entire inside of the unit with a mild detergent and soft cloth. Clean the treadmill belt as well.
8. Rinse out the unit.
9. Close the drain valve.
10. Fold the treadmill down to its normal position, install the treadmill guards and the foyer floor plates.
11. Refill the unit with fresh water. Equalized water depth is 31.5" or halfway up the top foyer door hinge.
12. Turn power back on to the unit.
13. Transfer water to the main chamber.
14. Follow water maintenance instructions to treat the water.

6.4 Treadmill Belt

The treadmill belt requires regular tracking and tensioning as the belt ages. There are tensioning screws located at the rear corners of the treadmill. Turning the screws clockwise while standing in the foyer will tighten the belt. To track the belt, if the belt has moved to the left than tighten the screw on the left part of a turn. Run the treadmill at 3mph for about 30 seconds and see if the belt has moved back to center. Make adjustments as needed to move the belt to center. If the belt moves too far than undo the last adjustment part way. The belt may stray slightly from center throughout the speed range and that is normal.

To check the treadmill belt tension, stand on the treadmill and turn it on at 2mph. If you notice any slippage then increase the belt tension using the screws located at the rear corners of the treadmill. Tighten each screw 1 turn and check the tension again until no slippage occurs. **NOTICE** Do not over tension the belt as that will cause the motor to overload and fault.

6.5 GFCI

The ECO is equipped with GFCI protection. Each of these devices shall be tested every month to verify they are operating correctly. **⚠ DANGER** **Risk of electric shock or death. If any of the devices are not operating correctly turn off power to the unit and do not operate**

until it is fixed by HydroWorx. There are 4 GFCI devices on the unit, 3 are located inside the load center (figure 6.5.1) and 1 is located beside the load center (figure 6.5.2).

To test the GFCI devices inside the load center press the small button in the center labeled test. Doing this should trip the breaker. If the breaker trips then the device is working properly, and you can reset the breaker. Repeat these steps for all three breakers.

To test the GFCI device located beside the load center press the test button. Doing this should make the device fault and the fault indicator light will illuminate. If the device faults than the device is working properly, and you can reset it by pressing the reset button.

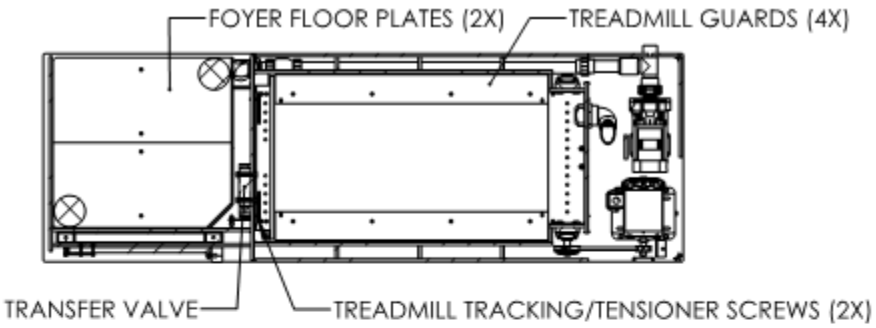


Figure 6.5.1

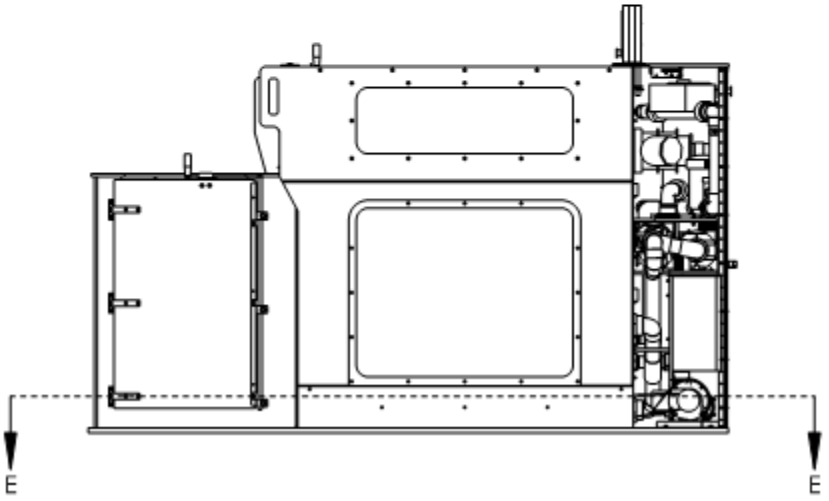


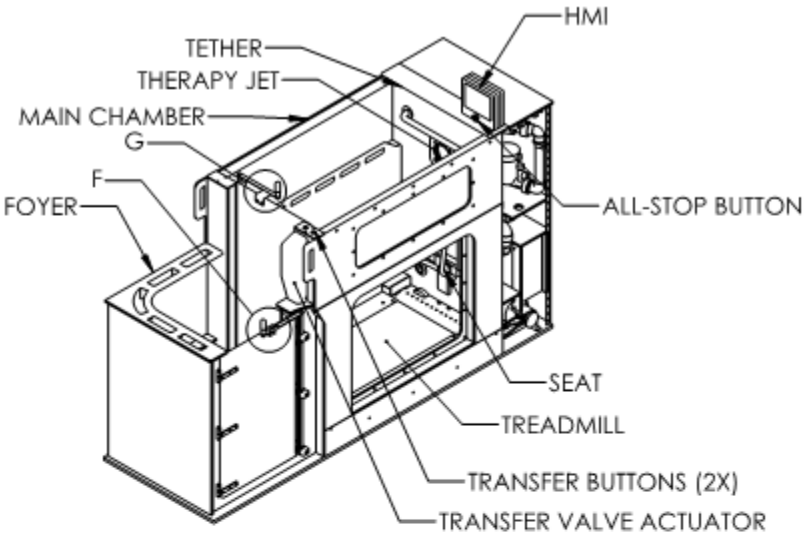
Figure 6.5.2

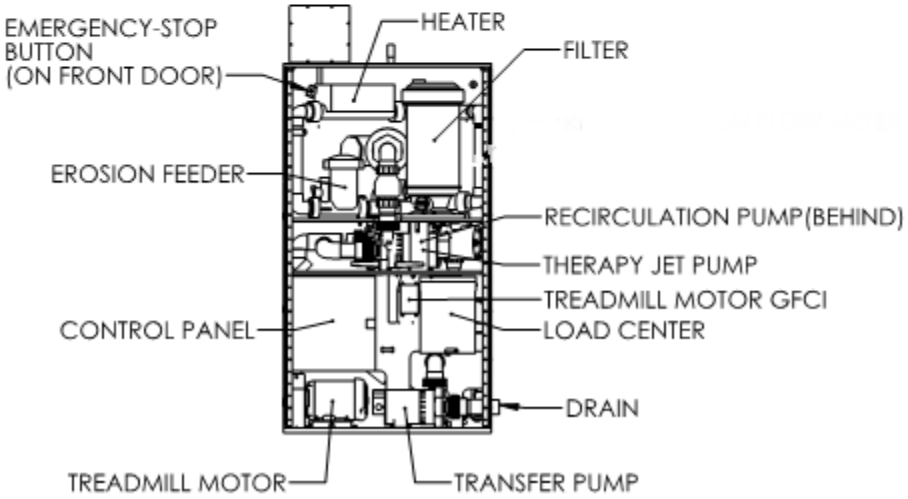
Appendix A – Drawings and Illustrations

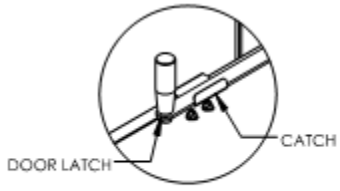


SECTION E-E
SCALE 1 : 24

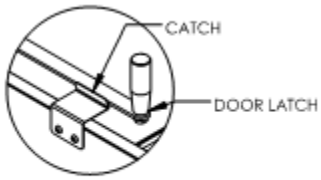








DETAIL F
SCALE 1 : 8



DETAIL G
SCALE 1 : 8